7.4.36

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Question: The abscissa of two points **A** and **B** are roots of the equation $x^2+2ax-b^2=0$ and their ordinates are roots of the equation $x^2+2px-q^2=0$. Find the equation and the radius of the circle with AB as diameter.

Solution

Define the Problem in Vector Form

The vector equation for a circle is given by:

$$\|\mathbf{x}\|^2 + 2\mathbf{u}^T\mathbf{x} + f = 0$$

where $\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}$, the center is $\mathbf{c} = -\mathbf{u}$, and $f = \|\mathbf{u}\|^2 - r^2$. Let the points be $\mathbf{x}_A = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$ and $\mathbf{x}_B = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$.

Determine the Vector **u**

Since AB is the diameter, the center $\mathbf{c}$ is the midpoint of $\mathbf{x}_A$ and $\mathbf{x}_B$.

$$\mathbf{c} = \frac{\mathbf{x}_A + \mathbf{x}_B}{2} = \frac{1}{2}\begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \end{pmatrix}$$

Using Vieta's formulas for the given quadratic equations:

- For $x^2 + 2ax - b^2 = 0$:

$$x_1 + x_2 = -2a$$
$$x_1x_2 = -b^2$$

- For $y^2 + 2py - q^2 = 0$:

$$y_1 + y_2 = -2p$$
$$y_1y_2 = -q^2$$

Substituting the sum of roots into the center vector $\mathbf{c}$:

$$\mathbf{c} = \frac{1}{2}\begin{pmatrix} -2a \\ -2p \end{pmatrix} = \begin{pmatrix} -a \\ -p \end{pmatrix}$$

Therefore, the vector $\mathbf{u}$ is:

$$\mathbf{u} = -\mathbf{c} = -\begin{pmatrix} -a \\ -p \end{pmatrix} = \begin{pmatrix} a \\ p \end{pmatrix}$$

Determine the Radius r and Scalar f

The squared diameter d^2 is $\|\mathbf{x}_A - \mathbf{x}_B\|^2$.

$$d^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$$

We use the identity $(m-n)^2 = (m+n)^2 - 4mn$.

$$\begin{aligned}(x_1 - x_2)^2 &= (x_1 + x_2)^2 - 4(x_1x_2) \\ &= (-2a)^2 - 4(-b^2) \\ &= 4a^2 + 4b^2 = 4\left(a^2 + b^2\right)\end{aligned}$$

$$\begin{aligned}(y_1 - y_2)^2 &= (y_1 + y_2)^2 - 4(y_1 y_2)\\ &= (-2p)^2 - 4(-q^2)\\ &= 4p^2 + 4q^2 = 4\left(p^2 + q^2\right)\end{aligned}$$

The squared diameter is:

$$d^2 = 4\left(a^2 + b^2\right) + 4\left(p^2 + q^2\right) = 4\left(a^2 + b^2 + p^2 + q^2\right)$$

The squared radius r^2 is $d^2/4$:

$$r^2 = a^2 + b^2 + p^2 + q^2$$

Now, we find the scalar $f = \|\mathbf{u}\| - r^2$.

$$\|\mathbf{u}\| = \left\|\begin{pmatrix} a \\ p \end{pmatrix}\right\|^2 = a^2 + p^2$$

$$\begin{aligned}f &= \left(a^2 + p^2\right) - \left(a^2 + b^2 + p^2 + q^2\right)\\ &= a^2 + p^2 - a^2 - b^2 - p^2 - q^2\\ &= -(b^2 + q^2)\end{aligned}$$

Final Equation and Radius Substitute $\mathbf{u}$ and f into the circle's vector equation:

$$\|\mathbf{x}\|^2 + 2\mathbf{u}^T\mathbf{x} + f = 0$$

$$\left\|\begin{pmatrix} x \\ y \end{pmatrix}\right\|^2 + 2\begin{pmatrix} a & p \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} - \left(b^2 + q^2\right) = 0$$

$$\left(x^2 + y^2\right) + 2\left(ax + py\right) - \left(b^2 + q^2\right) = 0$$

$$x^2 + y^2 + 2ax + 2py - b^2 - q^2 = 0$$

Equation of the circle:

$$x^2 + y^2 + 2ax + 2py - b^2 - q^2 = 0$$

Radius of the circle:

$$r = \sqrt{a^2 + b^2 + p^2 + q^2}$$